Yatranook

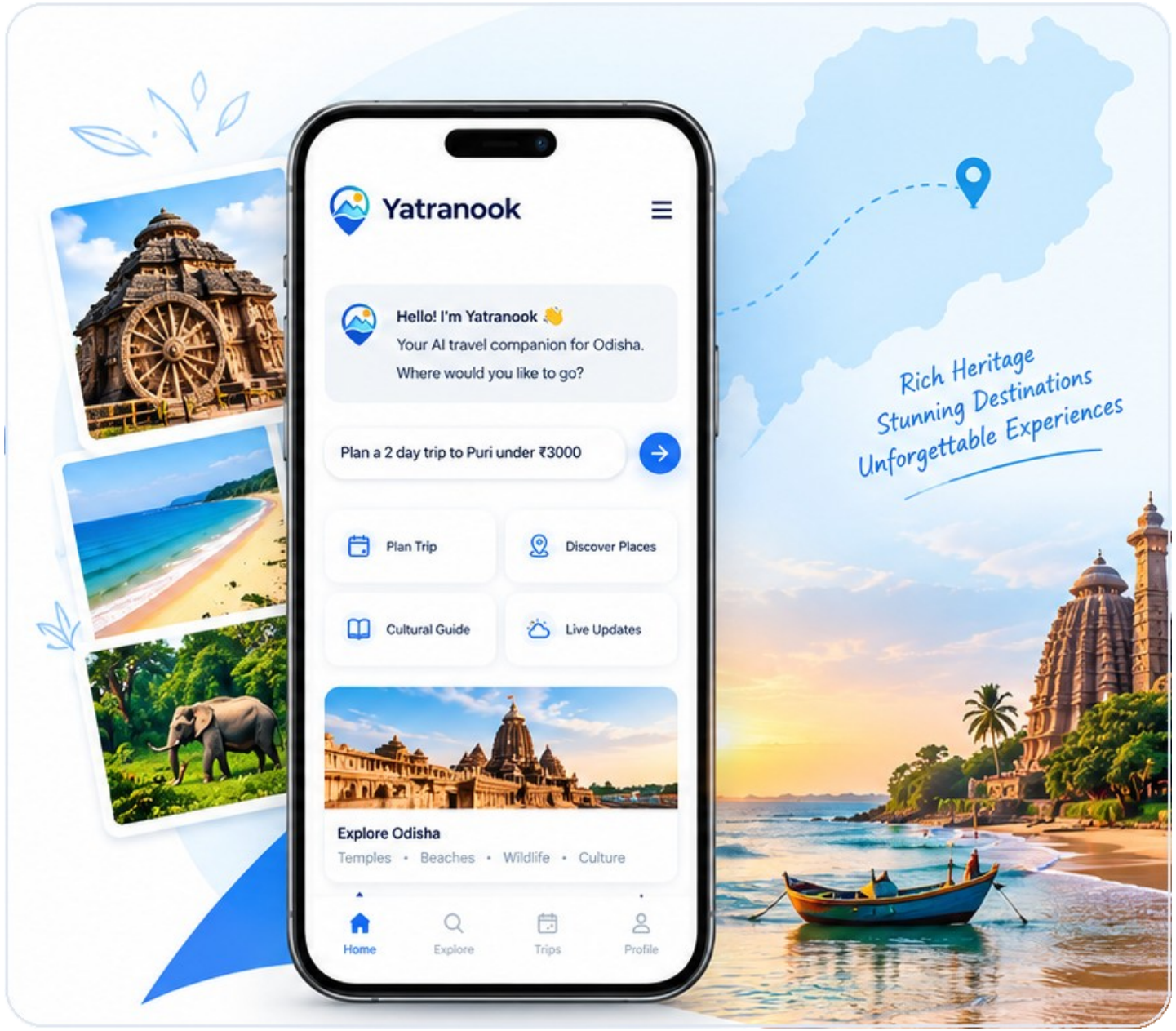
AI-Driven Intelligent Tourism Assistant for Odisha

Explore Odisha Smarter with Trusted AI

A source-aware travel companion that connects destination discovery, cultural knowledge and practical trip planning.

AI + TRAVEL TECHNOLOGY

RAG + PERSONALIZED ITINERARY PLANNING



Meet Yatranook: a smarter way to explore Odisha

An AI-powered tourism companion that connects destination discovery, cultural context and practical trip planning in one clear journey.

1 One place to begin

Explore destinations, experiences and regional highlights from a single starting point.

2 Trips shaped around you

Bring budget, dates, interests and pace together into a practical itinerary draft.

3 More than a list of places

Understand heritage, culture and local context with source-aware explanations.

4 Travel with more clarity

Make destination information easier to understand with language-aware assistance.

5 Clear about what is known

See sources and freshness cues; treat live data as optional and verify when needed.

6 Flexible when plans change

Revise the affected stops while keeping the rest of the trip constraints in view.

Our promise: bring discovery, cultural understanding and itinerary planning together—while showing assumptions, sources and uncertainty.

Where planning becomes difficult

Potential consequences include wasted time, unrealistic schedules, uncertainty and missed experiences.

01	02	03	04	05
Search	Compare	Plan	Travel	Recover
Multiple sources	Conflicting details	Constraints collide	Conditions change	Find an alternative

Opportunity	Keep trip context, explain recommendations, use evidence where available and revise plans without losing the traveler’s remaining constraints.

From fragmented search to a coherent travel workflow

Yatranook is designed to connect decisions across the journey, not simply generate chat responses.

TODAY: FRAGMENTED APPROACH

- Search in separate sites and social posts
- Manually combine route, time and cost
- Look up culture in separate references
- Re-check updates when plans change
- Verify whether answers have evidence

PROPOSED: Yatranook

- ✓ Ask in natural language with trip context
- ✓ Retrieve candidate destinations and evidence
- ✓ Generate a constraint-aware draft itinerary
- ✓ Refresh selected live data when available
- ✓ Show sources, freshness and uncertainty

The comparison describes a design goal. Baseline practices and improvements must be measured with users.

One assistant across discovery, understanding and planning

A source-aware interface connects traveler intent with structured destination data and practical constraints.

1 Understand

Conversational questions, trip context, cultural explanations and clarifying prompts.

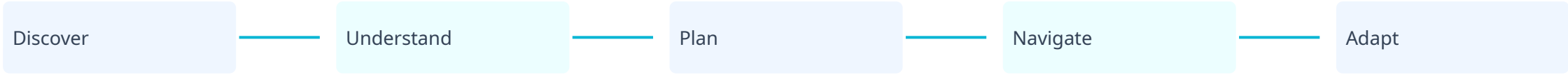
2 Plan

Personalized destination matching, day-wise itinerary drafts, budget and duration checks.

3 Trust

Retrieved evidence, source links, freshness labels, transparent estimates and uncertainty.

PROPOSED EXPERIENCE



Initial scope: responsive web prototype. Voice, advanced optimization and additional live integrations are later enhancements.

Design around distinct trip needs

Personas are intended research segments; priority order should be tested, not assumed.

1 Domestic visitors

Need practical multi-stop plans and clear costs.

Trip feasibility

2 International visitors

May need language help and cultural context.

Clarity + context

3 Families

Need manageable pacing and family-suitable options.

Comfort + logistics

4 Solo travelers

May value flexible discovery and reliable local information.

Confidence

5 Heritage enthusiasts

Want deeper explanations and respectful visit guidance.

Evidence + depth

Discovery questions: What is hardest to plan? Which facts must be trusted? How often do itineraries change?

Core features that work together

Capabilities are proposed; no feature is represented as implemented without code or test evidence.

1 **Conversational assistant**

Natural-language Q&A with trip context.

2 **Personalized itinerary**

Day-wise draft based on budget, time and interests.

3 **Destination discovery**

Interest, region and district filters.

4 **Cultural guide**

Source-backed history, architecture and etiquette.

5 **Map-aware planning**

Ordered places and provider-backed route estimates.

6 **Selected live info**

Weather, hours and transport only when reliable data exists.

7 **Multilingual support**

English, Odia and Hindi as priority languages.

8 **Source transparency**

Links, last-checked dates and uncertainty states.

9 **Adaptive updates**

Revise the plan while preserving remaining constraints.

A two-day Puri plan under a ₹3,000 budget

Illustrative only: actual prices, access rules, opening hours and route times must be checked for the selected dates.

TRIP REQUEST

“I have ₹3,000 and two days in Puri. Include temples, beaches and local food.”

Inputs captured

- Budget ceiling: ₹3,000
- Duration: 2 days
- Interests: heritage, coast, food
- Departure/stay details: ask if missing

DAY 1 | Puri

Morning: verify temple access and visitor guidance.
Afternoon: heritage lanes / local craft and food options.
Evening: Golden Beach, subject to conditions.

DAY 2 | Coast + heritage

Select a feasible nearby attraction such as Konark or a coastal stop only after checking transit, opening hours and remaining budget.

Illustrative budget allocation (not live quotes)

Stay	Food	Local travel	Activities	Buffer
₹900	₹600	₹700	₹400	₹400

Total = ₹3,000. Feasibility depends on travel-to-Puri costs, group size, season, accommodation and verified prices.

From request to an adaptive, evidence-informed plan

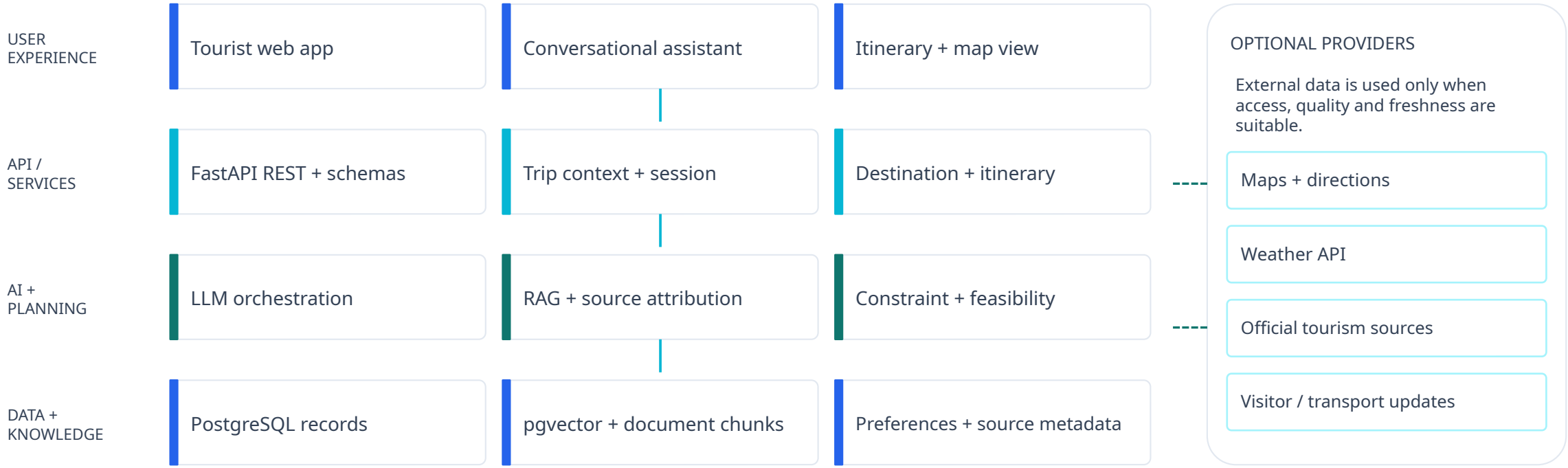
Decision gates prevent missing inputs or weak evidence from silently becoming confident answers.

1. **Collect**
   Location, dates, budget, duration, interests
2. **Clarify?**
   Ask questions if required inputs are missing
3. **Retrieve**
   Destination records + current data if available
4. **Draft**
   Recommendations with source-backed facts
5. **Validate**
   Budget, duration, routes and opening times where data exists
6. **Decision**
   If infeasible, revise or explain the trade-off
7. **Present**
   Plan + estimates + sources + uncertainties
8. **Adapt**
   Preserve constraints when plans change

Failure path: disclose unavailable data, avoid inventing facts, and offer a bounded alternative.

Architecture plan: modular services with evidence and validation

Core request flow is separated from optional providers and cross-cutting reliability controls.



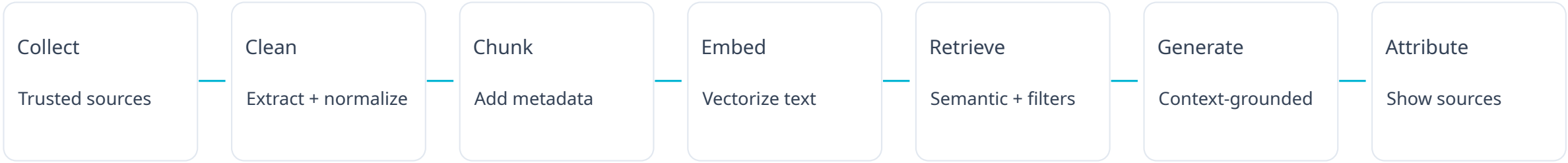
KNOWLEDGE PIPELINE — Trusted sources → clean + chunk → embed / index → attach metadata → review / refresh

CROSS-CUTTING CONTROLS — Source freshness + uncertainty • privacy / consent • provider timeouts • structured logs + tests + monitoring

Status: proposed reference architecture. Live integrations, production hosting and performance are not asserted as implemented.

RAG grounds responses in retrieved tourism evidence

Retrieval helps constrain answers but does not eliminate hallucinations or poor source quality.



Source metadata

Source title • URL • retrieval time • last verified • validity period • evidence status

When evidence is weak

Say what is unknown, avoid fabricated facts, and direct the user to a dependable source where appropriate.

Historical facts and live conditions need separate freshness policies.

Optimize usefulness inside hard travel constraints

Begin with transparent scoring and constraints; adopt more advanced optimization only if evidence warrants it.

INPUTS

Budget • duration • interests
start point • date • pace

CANDIDATES

Retrieve relevant places
with evidence + metadata

SCHEDULE

Estimate duration + travel
apply constraints + sequence

VALIDATE

Check known hours, time
and budget; flag unknowns

Candidate score (illustrative)

Score = w_1·interest match + w_2·location fit + w_3·heritage/nature preference − w_4·detour penalty

Hard constraints

- Total planned time ≤ trip duration
- Estimated spend ≤ stated budget
- Respect reliable opening / access limits
- Do not claim live route certainty without provider data

Weights require validation. Budget and travel-time estimates remain approximate and should carry provenance and uncertainty.

Trust has to be designed into the data and response flow

The system should communicate evidence boundaries rather than rely on confident-sounding language.

1 Evidence labels

Source title, URL and last-verified date where available.

2 Freshness tiers

Separate static cultural facts, reviewed records and live API responses.

3 Uncertainty states

Explicitly mark unavailable, stale or conflicting information.

4 Emergency guardrails

Display only verified emergency / official contacts; do not improvise.

5 Privacy by design

Collect only trip details needed for the interaction; define retention.

6 Operational controls

Input validation, logging, error handling and tested provider failures.

A pragmatic stack for a testable prototype

Treat all components below as architecture choices, not claims about an existing implementation.

Frontend	React + Tailwind CSS	Responsive web app, chat, search, itinerary cards and map view.
Backend	Python + FastAPI	REST endpoints, request schemas, service layer, errors and structured logs.
AI layer	LLM + embeddings + RAG	Retrieval-grounded answers, context assembly and source attribution.
Data	PostgreSQL + pgvector	Destination records, itinerary data, document chunks and metadata.
Integrations	Maps / directions + weather	Provider-backed live information when access and data quality allow.
Quality / delivery	Pytest + API tests + Docker + CI/CD	Test gates, environment-based secrets, monitoring and deployment readiness.

Build in phases with an acceptance gate at every step

No fixed delivery date is promised; timing depends on team size, data quality, provider access and scope.

01 Research + scope

Personas, validated needs, source policy

02 Foundation + data

API, destination schema, working search

03 RAG assistant

Curated evidence, retrieval, citations

04 Itinerary engine

Preferences, budget/time constraints, feasibility

05 Maps + live APIs

Routes/weather where supported, freshness states

06 Test + demo

Accuracy, constraints, failures, user evaluation

Release gate examples: searchable curated dataset → traceable answers → feasible draft trips → resilient demo with limitations disclosed.

Evaluate correctness, feasibility and user value

The project begins with a test plan; results should only be reported after controlled evaluation.

Factual accuracy	Correct claims / reviewed answer test set
Unsupported-claim rate	Answers containing unsupported factual claims
Constraint satisfaction	Trips meeting stated time / budget constraints
Travel-time error	Estimate vs route provider / observed travel time
Latency	Median and tail response time under test load
User satisfaction	Post-task ratings and interview themes
Citation correctness	Claim-to-source support reviewed by annotator
Recommendation relevance	Human relevance ratings for user scenarios
Cost estimate error	Estimated vs observed costs for comparable trips
API resilience	Timeout, empty and provider-error test cases
Language quality	Human review of English, Odia and Hindi outputs

Set thresholds before testing and publish sample size, evaluation method, date and known limitations with results.

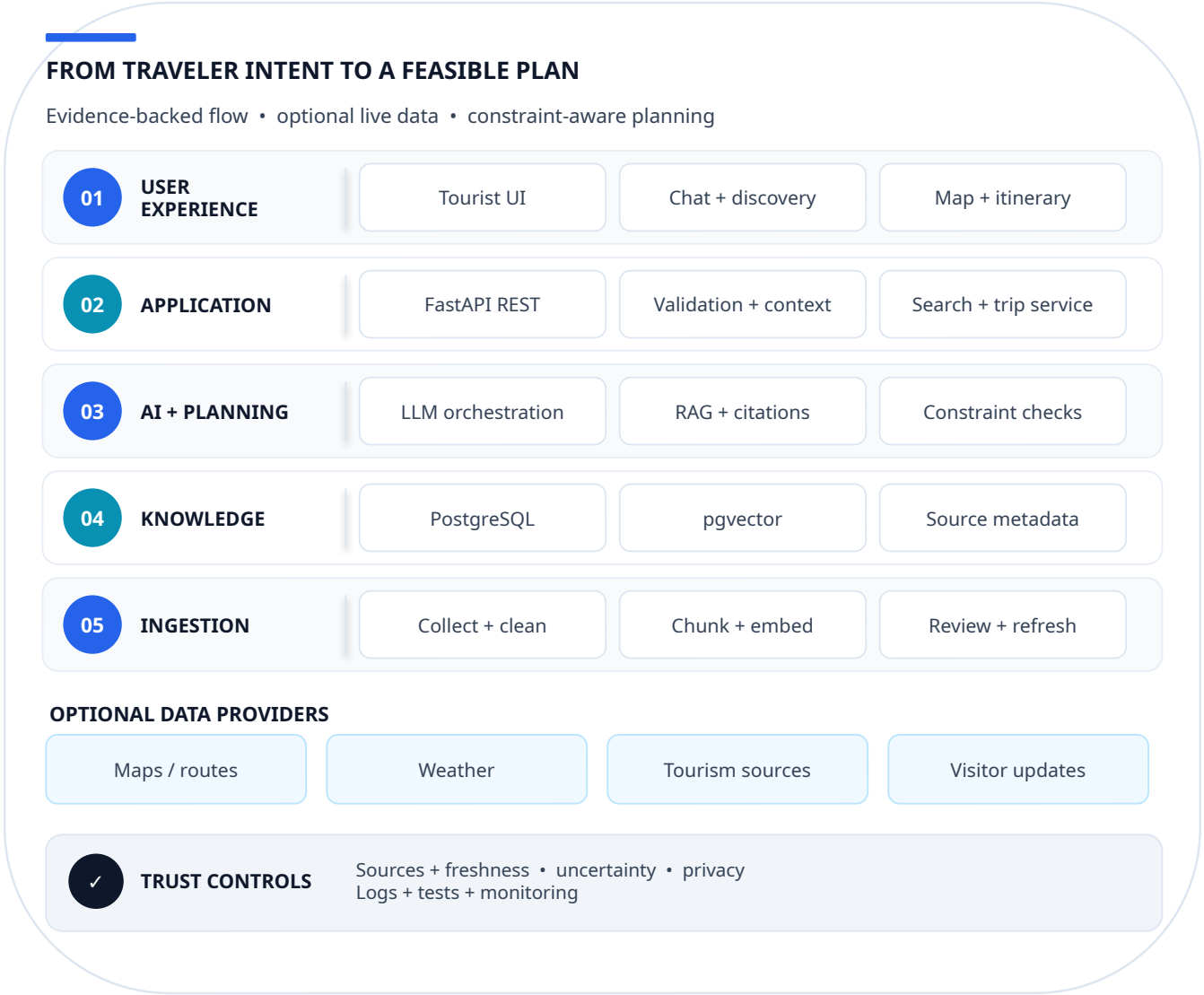
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NEXT STEP

Validate user needs → curate trusted data → test source-backed answers + itinerary feasibility.



Proposed concept • no claim of completed implementation, deployment or user validation.